
Hiking Trail Explorer — How To Run

Follow this checklist to get the Massachusetts hiking trail explorer running in minutes.

1. Install Prerequisites

1.1 Docker Desktop (includes Docker Compose)

1. Download the latest Windows installer from [Docker Desktop](#).
2. Run the installer and accept the default settings.
3. Launch Docker Desktop and wait until the status indicator reads **Running**.

💡 **Note:** PostGIS (PostgreSQL with spatial extensions) and Redis are provisioned automatically inside our Docker stack. You do not need to install them separately.

1.2 Windows PowerShell

You likely already have Windows PowerShell installed. We will use it to run the startup automation script.

1.3 Git (Recommended)

If you haven't installed Git yet, download it from git-scm.com to clone the project.

2. Clone and Open the Project

Open a PowerShell window and run the following commands to download the project and enter the directory:

```
PowerShell
# Clone the repository
git clone https://github.com/yundatwit/Hiking-Trail-API.git

# Enter the project folder
cd Hiking-Trail-API
```

3. Obtain API Keys (Required)

To make the map and weather features work, you need two free API keys.

3.1 Google Maps Platform

1. Visit the [Google Cloud Console](#).
2. Create a new project (or select an existing one).

3. Enable the following APIs:
 - **Maps JavaScript API**
 - **Geocoding API**
4. Create an API Key ([Credentials](#) > [Create Credentials](#) > [API key](#)).
5. Copy this key—you will need it for the `.env` file.

3.2 WeatherStack

1. Sign up for a free account at [WeatherStack](#).
2. From your dashboard, copy your **API Access Key**.

4. Prepare the Configuration

1. While inside the `Hiking-Trail-API` folder in PowerShell, check if an `.env` file exists. If not, create one from the example:
2. PowerShell

```
Copy-Item .env.example .env
```

- 3.
4. Open the `.env` file in Notepad (or your preferred code editor):
5. PowerShell

```
notepad .env
```

- 6.
7. Replace the placeholder text with your actual keys:
 - `Maps_KEY=your_google_maps_key_here`
 - `WEATHERSTACK_KEY=your_weatherstack_key_here`
8. Save and close the file.

5. Run the Automated Startup Script

Ensure Docker Desktop is running, then execute the startup script from the project root:

PowerShell

```
powershell -ExecutionPolicy Bypass -File .\start.ps1
```

What this script does:

- Verifies Docker is running.
- Checks for your `.env` file.
- Builds and starts all Docker services (Backend, Frontend, Database, Redis).
- Runs database migrations.
- Seeds the database with Massachusetts trail data.
- Automatically opens <http://localhost:3000> in your default browser.

*Note: Stay in the PowerShell window until you see the success banner. You can press **Ctrl+C** to stop viewing logs when finished.*

6. Verify the Application

1. If the browser didn't open automatically, navigate to:
`http://localhost:3000`
2. You should see the **Trail Finder Dashboard**, including:
 - Search bar for cities.
 - Interactive Google Map.
 - Current Weather panel.
3. (Optional) Test the backend API directly:
4. PowerShell

```
curl "http://localhost:5000/api/trails/search?city=Boston"
```

5.

7. Useful Commands

Action	Command
View Live Logs	<code>docker-compose logs -f</code>
Stop Application	<code>docker-compose down</code>
Rebuild After Code Changes	<code>docker-compose down</code> <code>docker-compose up -d --build</code>
Reset Database (Delete Data)	<code>docker-compose down -v</code>

	<code>docker-compose up -d --build</code>
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8. Troubleshooting

- **Blank Map:** If the map loads but shows a gray screen or "Development Purposes Only" watermark, verify that your **Google Maps API Key** is valid and has billing enabled (even for the free tier).
- **Missing Weather:** Check your **WeatherStack** usage limits or ensure the key is copied correctly in the `.env` file.
- **Docker Errors:** Ensure no other services (like a local Postgres server) are using ports `5432` or `3000`.

 **You're all set!** The Hiking Trail Explorer should now be fully operational on your local machine.